Using Possible World-States to Understand

The Liar Paradox

Intellectual Property of Justin Edward Oxford

Justin Edward Oxford

17 December 2018

Professor Richard K. Atkins
Truth & Pragmatism
Boston College

The Liar Paradox has plagued the minds of philosophers since the time of the ancient Greeks. Many great minds have toiled over the contradiction, and equally many solutions have been thrown against the wall: invalidation of self-referential language, semantic ascent, deflation, logical proof, norms of assertion, and so on. Each of these solutions has its drawbacks and advantages – and each has plenty of prerequisite acceptances. But just what is the Liar Paradox? It can be stated as follows:

A: The sentence A is false.

Right, that's well and good, is it not? One could take the side of apathy – many often do when the paradox is brought up[1]. However, if the claim is inspected closely, a problem emerges. Let us assume that A (the whole phrase written above) is true. Then we have the claim that, "The sentence A is false" is true. Therefore, the sentence A is false, but we just assumed that it is true. This is a contradiction! Fine, assume that it is false. Now we have, "The sentence A is false" is false. Therefore, the sentence A must be true (since true is the opposite of false, and false means that the opposite is true). But we again have an issue. We assumed A was false, but have now concluded that it is true – another contradiction! In fact, we can carry on *ad infinitum* by tacking on our newly derived truth value: """"The statement A is false", is true", is false", is… herein lies the paradox. The sentence appears to be simultaneously true and false.

The Liar Paradox may seem as if it is merely a one-off exception of little consequence to our understanding of truth. However, there is no mention of a theory of truth in the explanation above. The issue arises from our common sense use of truth alone, and while it is an extreme case, there is no guarantee that the rest of our conception of truth is free from gaps. The question arises of whether we should accept or deny this utterance, A, as a valid use of language or occurrence of thought. Clearly it follows grammatical syntax for the language in which it is spoken, but it also appears to be complete nonsense and meaningless. In this paper I shall argue for the later. In order to understand the linguistic aspect of the paradox, we will examine the solution proposed by Alfred Tarski. We then will move to understand C.S. Peirce's solution of self-reference, and finally, we will resolve to the correct solution.

I. Alfred Tarski
A Semantic Solution to the Paradox

Alfred Tarski proposed a solution to the Liar Paradox wherein he claims that language exists on multiple semantic levels, and it is the ascent through these levels that supposedly solves the paradox. According to Tarski, there are four basic semantic levels in the concept of language: things themselves, words, an object-language, and meta-language(s). Things themselves are entities which exist somewhere in the universe – they are the actual, physical, real, tangible bit of the universe. Things themselves are the base layer – what most would simply call Reality. Words are language symbols for things themselves. A coffee maker exists independently of language – it is physically real, tangible, etc. – but it is not *innately* a coffee maker. 'Coffee Maker' is the vocal-vibration pattern, mouth and throat shape, and written symbolization of that physical object. Words have definitions in terms of the description of the object they symbolize. For example, 'snow' has

[1] "*Portal 2: GLaDOS Paradox*": https://www.youtube.com/watch?v=JR4H76SCCzY

Intellectual Property of Gavin Edward Oxford

numerous physical (or real) properties – i.e. it is white[2], it is in a certain temperature range, it is 'flaky'[3], etc. – thus we define the word 'snow' as being of and having those properties.

The Object-Language and Meta-Language(s) are where the actual structure of language exists. They allow us to move up a level from merely running around pointing at things and saying what they are to relating things to each other, commenting on those relationships, and constructing full thoughts. As we ascend the semantic ladder, Tarski proposes that the first language we encounter is the Object-Language. This is the language of statements about the world – i.e. "Snow is white", "Horse is bay", "Coffee is hot". In the Object-Language we are picking out aspects of the world and stating them as they relate to one another. However, at this level we are still one step removed from the truth. As Colin McGinn says in *Tarski's Theory of Truth,* "The object language is the language we are speaking about when we formulate our definition of truth for that language" (156).[4] By this, McGinn means that truth does not exist on this semantic level, at least in the conceptual sense. Instead, this language is what is questioned when questions of truth or falsity arise. To deal with these questions, Tarski proposes a fourth level. He calls this the Meta-Language. "The [Meta-Language] … is the language that we are using to talk about another language" (156).[3] This other language in general is the Object-Language, but it can also be lower-level Meta-Languages.

It is only in the Meta-Language that questions of truth or falsity are validly asked. It is not possible to say, "X is true" in the Object-Language because, as is implied by Tarski's theory, truth does not exist in reality. Truth for Tarski is one of the ways in which a Meta-Language talks about an Object-Language. "Snow is white" contains no mention of truth – although, as we will see, Peirce questions whether this is the case when he addresses self-assertion of truth of propositions. "Snow is white" is an Object-Language Proposition which claims that the world is a certain way. Truth enters the scene when the validity, verification, or accuracy of the proposition is brought into question. It is at this point where we ask questions and make statements *about* the Object-Language Proposition. This is the ascent into the Meta-Language: "Snow is white" is true.

When we enumerate the levels of ascent, we begin by labeling the Object-Language as L_0, the first step into the Meta-Language as L_1, the next Meta-Language as L_2, and so on *ad infinitum* or as far as we wish to go. It is important to note, for clarity's sake, that *the Object-Language-per-se of L_2 is L_1,* and so on. Additionally, we can include the reality/things themselves level and the word-level to produce a better picture of what is occurring. Here is how this listing appears in the case of "Snow is white":

R: *snow* / *white*[5]
W: "snow" / "white"[6]
L_0: "Snow is white."
L_1: "Snow is white" is true.
L_2: "'Snow is white' is true" is true.
⋮

[2] This is also a *word* which represents a reality, namely, the color white.
[3] I studied philosophy, not atmospheric science.
[4] McGinn, Colin. "*Tarski's Theory of Truth*." In Philosophy of Language: The Classics Explained, p.147-64. MIT Press, 2015. http://www.jstor.org.proxy.bc.edu/stable/j.ctt17kk9zd.11.
[5] This is the base-level – things themselves – wherein * denotes the *thing itself*.
[6] This is the word-level wherein words in a language are used to symbolize things themselves.

Intellectual Property of Justin Hyman ©2017

L_n: "L_{n-1}" is true.

The enumeration of these levels of ascent is paramount in Tarski's theory. "Tarski insists that paradox will be inevitable if one takes both [L_0] and [L_1] to be the whole of some natural language, including the word "true" itself" (18).[7] That is, we must not assume that two different levels of ascent are referencing truth or falsity in the same language. L_0 and L_1 neither reference or exist in English (or human language) as a whole, rather they exist in their own respective languages. Instead, if we recognize that the Meta-Language is commenting on the Object-Language in reference to the Object-Language itself, then we eliminate the contradiction. This is the solution at the heart of Tarski's theory: when we talk of the truth of X, we are talking of the truth of X *in its respective language*. As summarized by Burgess & Burgess: "If, however, one distinguishes the object language [L_0] from the metalanguage [L_1], and takes 'true sentence of [L_0]' to belong only to the latter and not the former, then," (19) we have:[8]

L_0: "Snow is white."
L_1: "Snow is white" is a true *sentence in L_0*.
L_2: "'Snow is white' is true sentence in L_0" is a true sentence in L_1.
⋮
L_n: "L_{n-1}" is a true sentence in L_{n-1}.

Thus is Tarski's semantic theory of truth. Let us now consider what occurs when we insert the Liar Paradox into this system. The first note to make is that we can no longer simply say: "This sentence is false." According to Tarski, X is true or false as a sentence in a language. Thus, we append the language classification to the sentence as follows: "This sentence in false as a sentence in L_0." It is here that an important note must be made. As I said previously, Meta-Language sentences may reference other Meta-Language sentences. This is to say that an Object-Language sentence is not a necessary condition to derive a Meta-Language sentence. Instead, all that is needed is the desire to speak about some sentence and assign it a quality. Let us now turn to Tarski's solution.

Assume there is an utterance:

(1) "(1) is false."

According Tarski, assertions about qualities of statements (especially those of truth or falsity) must make reference to some semantic level. Assume that (1) references a statement in a semantic level, L_n. Then we have the correction:

(2) "(2) is false in L_n."

Now, assume that we wish to assert that (2) is true. We must ascend to a higher Meta-Language, L_{n+1}, and thus we must say that (2) is true in L_{n+1}. By Tarski, we have:

(3) "(2) is false in L_n." is true in L_{n+1}.

Further, consider the T-Schema, (T), which states:

[7] Burgess, Alexis G. & John P. "*Truth*." p.18. Princeton University Press, 2011.
[8] Burgess. *Truth*, 19.

Intellectual Property of Justin Edward Oxford

(T) "P" is true in L_x iff P

Thus,

(4) "(2) is false in L_n." is true in L_{n+1} iff "(2) is false in L_n."

Lastly, by claiming that, (2) is "(2) is false in L_n", and substituting appropriately, we arrive at the following:

(5) (2) is true in L_{n+1} iff (2) is false in L_n.

Behold! There is no contradiction. (2) is both claimed to be true and claimed to be false, but these claims are occurring on different semantic levels. Because of this fact, there is no infinite regress and no bouncing back and forth between true and false along the way. Moreover, no matter which further (and valid) substitutions are made, there will be no contradiction as the Meta-Language assignments will always be different. The only way this statement is rendered contradictory is via the elimination of the Meta-Language assignments. Without the levels of ascent denoted, one is left with: "(2) is true iff (2) is false." This is a clear contradiction, but according to Tarski, it is a contradiction only insofar as it is a misrepresentation of what is actually occurring within the language. However, there are some issues which Tarski fails to address.

As with most famous problems in Philosophy, there is a great deal of discussion surrounding the Liar Paradox, and there has been for some time. There are a multitude of competing theories and straight rejections of Tarski's seemingly bulletproof logical solution. While the above does appear to solve the issue on the surface, this efficacy is only skin deep. I am inclined to agree with those critics who oppose Tarski's solution. There are issues such as: statements which reference all semantic levels being unable appear in the list without including themselves, issue of normativity and use of "True-in-L_n" in daily conversation, and its incompatibility with what I consider to be a more robust and accurate theory of truth. In the pages that follow I will aim to explicate these issues in terms of Tarski, evaluate them, and begin to point towards a better solution to the Liar Paradox.

The first issue in need of treatment is the issue of what I will call Universal Claims of Ascent.[9] If we assume that Tarski's solution is the proper method for handling the paradox, we find that it is possible to work around his rules while also remaining within them. For example, a statement which addresses every possible element of Tarski's ascent ladder would have to exist above the rest of the languages it is addressing, but if it is addressing every language then it must also contain itself or exist above them – it cannot be both. Consider the following claim: "For every statement in [L_n] of the hierarchy, there is a statement at [L_{n+1}] which asserts that the first statement is false."[10] As this is a claim about every possible L_n, it must reside in a Meta-Physical Language above all other levels of ascent. However, it also claims that there is a level above itself. If this is true, then Tarski's solution to the paradox creates yet another paradox. Tarski could possibly defend himself by creating a fifth language – call it a Systems-Language – which is the moving of

[9] I very much wish to cite this view as I did not come up with it on my own. However, it is stated on the Wikipedia page as a possible objection to Tarski, and I am unable to find the source as it is listed with [Citation Needed]. This is a shame as I would have liked to track down the author for further reading. Therefore, I will direct the reader to the Liar Paradox Wikipedia page in hopes that a reference is added and credit is properly given. Moreover, I acknowledge the poor practice of citing Wikipedia, but in this case I feel an exception is warranted as the objection raised is rather sound and is worthy of consideration.

[10] See footnote 9.

Intellectual Property of Justin Edward Oxford

perspective *from* the Meta-Languages to the *viewing of* the Meta-Languages themselves (i.e. words describe things, Object-Language uses and describes words, Meta-Language describes the Object-Language, and Systems-Language describes the Meta-Language). However, it is then possible, by Tarski's own line of thought, to ascend through the Systems-Language in the manner similar to his ascent in the Meta-Language. Once again we could further remove our perspective *from* the Systems-Language to some other language. All this would do is move the problem to some other language.

Perhaps this fluke does not matter. Having studied Mathematics in some depth, I have learned to accept that at infinity things become quite weird. Perhaps the claim above exists right at the limit; perhaps it is in the *theoretical* last propositional level. Then there is an issue far more fatal. If the claim, let us call it U, exists in the theoretical last level, then it is false because it states that by definition there is a level above it. But by the very utterance of this falsity we have shown that there is indeed a level above U – the one in which we are talking at this very moment. If U is the in the last propositional level, we are left in a paradox where it appears to be both the last and the not-last. But if U is not in the last propositional level, we appear to be breaking the system which Tarski designed to explain semantic ascent in that U references all Meta-Language levels yet does not appear above all levels in references. Thus, there is no conceivable way in which to solve this issue given Tarski's system – every angle leads to further paradox.

The second problems facing Tarski's analysis is the issue of normativity. We do not use "True-in-L_n" in daily conversations about truth. When Phil says, "Snow is white," and I say, "What Phil said is true," I do not point out or conceptualize that I am semantically ascending. Moreover, I do no mean to be saying anything about the proposition, "Snow is white," at all. Instead, I am agreeing with Phil and asserting that what he said maps onto the world in an exact way; the essence of his statement's content is verifiable to the way the world actually is. I am not analyzing his proposition.

It is at this point that one must depart and distinguish one's views from the deflationary views of truth. The use of "is true" is undeniably more than a semantic tool which one uses to address propositions. The predicate "is true" aims at the content of a proposition and its relation to the world and to reality. Although it can appear to function in a deflationary, semantic way and in few cases may even do so, it is the nature of truth that it is linked to physical reality – it 'gets at the world'. The proposition P is the icing on top of a birthday cake that reads, "This cake is delicious." I say that P is true because of what is underneath, that is the object of my conceptualization of the truth, not the proposition itself. Semantic-only conceptualizations of truth are a subset of all possible conceptualizations of the concept, and they do not exhaust the set – they are indeed a small portion.

Putting aside the deflationary theories, and going back to Tarski, it is clear that in the system Tarski imagines, his solution works. However, his imagination is not an accurate account of what is occurring when one says, "…is true." This is the case for normative and syntactical reasons.

Lastly, Tarski's solution to the Liar Paradox assumes that Truth is a semantic notion. By this, I understand him to mean that he treats truth as a logical value, and nothing more. For Tarski, truth is just a number in a balanced equation – it can be passed around *willy-nilly,* added or removed, and the only thing it adds is reference to some other proposition. However, the line of thought above must be continued. By appending the predicate, "…is true," onto a proposition, one

Intellectual Property of Justin Edward Oxford

is saying far more than a mere description of the qualities of the proposition that one references. There is a pragmatic element to truth which surpasses language. Truth would exist without language. The liar paradox would exist without language – I am able to think, *I did not just think this,*[11] and we arrive at a near identical conundrum. While one is unable to express or transfer thoughts to others without language – at least until humanity catches up to science-fiction – thoughts and conceptualizations themselves to not require language. Thus, Tarski's attempt to conceptualize truth in terms of language is flawed from the outset.

Consider Tarski's underlying claim that truth can only exist in language alone, and let us assume this to be the correct account. It is therefore apparent that since language is the expression or transfer of thoughts between minds, and in some circumstance one wishes to agree with Jones that *such-and-such is true*, then one necessarily needs language to agree with Jones that *such-and-such is true*, and their entire purpose in saying this is to convey that thought. But this is not truth simpliciter; this is *agreement* of truth. In Tarski's model, one is not making any other reference other than my reference to the proposition. But this is a nonsensical way to view truth. By viewing truth in this way, a language which does not remotely reference the world can contain truth or falsity. I am certain that the vast majority of people would say that running around making claims about propositions is not 'getting at the truth' – it is getting at non-referential nonsense.

Moreover, Truth can exist in one's own mind and is not merely an expression of, or transfer of the thought of, agreement. Expressivist notions of truth are flawed for this reason. Truth surpasses language almost instantly, and language is only needed to convey *that* we find something true *once* we have found it to be so. Thus, one's agreeing with Jones that such-and-such is true occurs only after one has left the lingual-propositional level, journeyed to their concepts of reality, and returned.

I claimed above that truth exists independent of language. How might truth appear in thought alone, without language? Consider the following situation. I am trapped, alone on a desert island, and prior to my arrival my entire language faculty had been wiped away. Would I still understand matters of truth and falsity? Yes. I could think: *I ate that berry, and it made me sick. I am beginning to think that: those berries are not something that I should eat. I wonder if that is true, so I will eat another to see – maybe it was just the one that was rotten*. These thoughts exist prior to language – presumably, just as they existed to our ancestors before they had language.[12] This is the process of inquiry, and it is within this context that truth resides.

In all of this, I simply aim to say that truth is not language dependent. Indeed, truth exists beyond language, and Tarski's solution of stopping his conception of truth at the linguistic – or semantic – level is a mistake.

In summary, while Tarski's theory appears to be a valid solution to the Liar Paradox, and it indeed solves the riddle within its own formulation of the rules of truth and language, it ultimately comes up short. There are numerous problems and no apparent solution[13] to them. Tarski fails to consider a paradox which his own constructed system creates when one makes

[11] I am using, *, here to denote thoughts.

[12] Further, if one accepts language to have been socially constructed – or at least to have not existed at some point – it necessarily follows that thought precedes language and can exist without it. And truth certainly existed then as it does now.

[13] Within Tarski's rules and concepts – there are certainly solutions to the greater problems.

Intellectual Property of Griffin Edwards Oxford

Universal Claims of Ascent. His solution does not take into account normative uses of truth or natural conversational language. By extension, Tarski fails to present an accurate account of truth's relation to language and thought, further disconnecting his theory from normative conception. For these reasons, Tarski must be rejected.

■■■

II. Charles Sanders Peirce

Assertion, Assignment of Meaning, and Invalidity of the Paradox

One interesting characteristic of the Liar Paradox is that is it self-referential – "*This statement* is false." Thus, the first solution one may be tempted to propose is to merely disallow self-reference. That is to say, "All self-referencing statements are invalid." And if one does this, it eliminates the paradox. Further, it explains that the reason for the paradox is self-reference specifically. However, this is not a valid, general solution. There are self-referential claims which do not end in paradox such as, "This sentence is in English," and, "This sentence has five words." Both of these sentences are true – if we assume them to be true, we find that it is so, and if we assume them to be false, we find that we are in error. There is something different going on within the paradox itself, and it has less to do with the subject, "This proposition," and more to do with the predicate, "…is true." However, the philosopher Charles Sanders Peirce felt that the paradox could be tackled from the front.

C. S. Peirce initially attacked this side of the liar paradox in 1869 in his essay *Grounds of Validity of the Laws of Logic: Further Consequences of Four Incapacities*. He focused on the proposition itself, claiming that every proposition implicitly begins (whether it is written or not), "This proposition…". That is, all propositions say something about themselves implicitly. Further, that which they say about themselves is that they are true. As Richard Atkins points out in his essay *This Proposition is Not True: C.S. Peirce and the Liar Paradox*, Peirce's aim was to show that: "Since every proposition implicitly asserts its own truth and [The Liar Paradox] explicitly asserts it is not true, [The Liar Paradox] both asserts it is true (implicitly) and asserts it is not true (explicitly). That is self-contradictory. Self-contradictory propositions are false. Hence, [The Liar Paradox] is false" (424).[14] Further, as Atkins says, "[Peirce's] solution hinges upon the claim that *every proposition asserts its own truth*" (424).[15] By accepting Peirce's claim, the liar paradox is resolved to a single value: false. But one is left with a further explanation in regards to the assumption of self-assertion-of-truth. However, the acceptance of self-assertion does not appear to be Peirce's ultimate view.

In 1903, C. S. Peirce gave a series of lectures at Harvard University. In his third lecture, Peirce resumes his line of thought from 1869. He begins by saying, "Take, then, the question of whether a judgment judges itself to be true. Unquestionably if one judgment does so, every

[14] Atkins, Richard K. "*This Proposition is Not True: C.S. Peirce and the Liar Paradox".* In Transactions of the Charles S. Peirce Society: A Quarterly Journal in American Philosophy, Volume 47, Number 4, pp. 424. Indiana University Press, 2011

[15] Ibid. pp. 424.

Intellectual property of Justin Harvard Oxford

judgment does so" (166).[16] Under his 1869 conception, this matter in question is certainly the case. However, Peirce moves to reject his previous notion. He aims to say that propositions cannot and do not assert their own truth. Peirce rejects his previous conception for three reasons: circularity in his reasoning, common sense, and the invalidity of self-reference. The last of which will prove vital to our final solution.

In the first of these rejections, Peirce claims that his reasoning is circular - *petitio principii*[17]. "His proof that [The Liar Paradox] is self-contradictory depends upon the premise that every proposition implicitly asserts its own truth" (424).[18] This was Peirce's solution, as mentioned above:

> "This proposition is not true… is certainly a proposition. Hence by the definition of a proposition it is either true or not. But suppose it involves no falsity. Then it will follow that it is not true and does involve falsity, thus reducing that hypothesis to absurdity. The proposition is therefore not only false but absurd and self-contradictory. But all that it *explicitly* asserts is that it is not true" (166-67).

Therefore, Peirce claims, it must follow that a proposition *implicitly* asserts its own truth – otherwise we would not have arrived at a contradiction.

"However, his proof that every proposition implicitly asserts its own truth depends upon the assumption that the paradoxical nature of [The Liar Paradox] arises from an inconsistency in the nature of the proposition itself" (424).[19] Peirce says, "Consequently, the only way in which it can involve contradiction is by expressing at the same time the assertion that it is true…" - this is Peirce's first conclusion – "We must therefore conclude that every proposition, in the very propositional form itself, expresses the assertion of its own truth" (167).[20] These conclusion are circular. Thus, Peirce concludes that his initial claim was invalid.

Peirce's second rejection is far less glamorous or rigorous, thus we will spend little time entertaining it. His example is of a court case where all evidence is lost except for two testimonies. "A testifies that B's testimony is true," and, "B testifies that A's testimony is false… Common sense would certainly declare that nothing whatever was [asserted]" (167).[21] Thus, if in this contradictory case we agree that nothing was asserted (or testified) then it follows that the liar paradox does not assert anything either – certainly not its own truth.

Third, Peirce addresses the question of when exactly a proposition is to have asserted something. He does this by first making the claim that the subject term of a proposition is aimed at an object. This object exists in the world somewhere prior to the stating of the proposition. This is consistent with the rebuttal of Tarski above wherein it was argued that language is not prior to thought, for thought comes from reality itself and our perception of it. Thought is then manifested in language to be transferred between persons or minds.

Intellectual Property of Justin Edward Oxford

---

[16] The Peirce Edition Project, Andre De Tienne, Jonathan R. Eller, Albert C. Lewis, Cathy L. Clark, and D. Bront Davis, eds. *The Essential Peirce: Selected Philosophical Writings (1893-1913)*. BLOOMINGTON; INDIANAPOLIS: Indiana University Press, 1998. http://www.jstor.org.proxy.bc.edu/stable/j.ctt16gz4vr.
[17] Ibid. pp. 167.
[18] Atkins. pp. 424.
[19] Ibid. pp. 424.
[20] The Pierce Edition Project. pp. 167.
[21] Ibid. pp. 167.

This aim at an object, says Piece, is done via indices. An index is an indication of an object in the world.[22] Thus, a proposition *indicates* where an object is to be found – in thought, time, space, etc. Peirce then says:

> "Consequently, though the subject need not be individual, the object to which the subject of a proposition applies must be the object of a possible index and as such it must be such as it is independently of any representamen or other Third. That is to say it must be *real*… Consequently, it is impossible that a proposition should relate to itself as its object, since as long as it has not yet been enunciated it possesses characters which are not independent of how they may be represented to be. It is, therefore, quite impossible that a proposition should assert its own truth, or what comes to the same thing, that a desire should desire its own gratification, or that an argument should conclude its own cogency, excepting only in that sense in which a point may map itself to itself, namely, as a special case under a general representation" (168-69).[23]

This is to say that until a proposition has been enunciated, it is not a *real* object, and thus it does not have a possible index. Therefore, it cannot be referenced. However, this appears to be contradictory to the belief that we can have propositions such as, "This sentence has five words." That is, it appears to be the denial of the validity of self-reference all over again. Richard Atkins has a different understanding of what exactly Peirce is saying in the above passage. He says:

> "I do not think that the sentence beginning '*Consequently, it is impossible that a proposition should relate to itself as its object*' should be read as affirming that propositions cannot relate to themselves at all… To the contrary, Peirce importantly qualifies his point: '*since as long as it has not been enunciated it possesses characters which are not independent of how they may be represented to be*.' The key point here is that truth cannot be asserted of a proposition until the proposition has been enunciated. Once it has been enunciated, truth can be asserted of it by a rational agent... Thus, propositions cannot assert their own truth because they have not yet been enunciated" (427-28).[24]

Atkin's assertion turns on the reading of the word *characters* as being qualities specifically of truth, falsity, opinion, etc. Further, it assumes qualities such as being-in-English and having-five-words are not such characters. However, it seems as if Peirce is confused, and rather than a qualified understanding, Atkins offers a correction to Peirce's view. There is nothing in this literature that I can understand to not include *all* self-referential statements. The phrase, "*Consequently, it is impossible that a proposition should relate to itself as its object*," is inconsistent with what follows, "*since as long as it has not been enunciated it possesses characters which are not independent of how they may be represented to be*." In the latter portion of Peirce's claim, he implies that a quality, such as being-in-English, is not an independent quality, but it certainly is. Atkins is correct in his assessment of what the proper understanding should be, but this is not the view Peirce puts forth. A proposition cannot reference something which is not able to be indexed or is not an object, but it is not self-reference that causes the issue to arise. Instead, it is something hidden deeper within the Liar's statement that requires a different understanding of truth.

Therefore, while Peirce had begun to get at the heart of the issue, he was not able to fully flesh out the nature of the paradox. Richard Atkins correctly amended Peirce's work in his essay, and Atkins offers further solution which I have chosen not to treat in this essay as I believe that

[22] "Now to INDICATE is to represent in the manner in which an index represents." Ibid. pp. 168.
[23] Ibid. pp. 168-69.
[24] Atkins. pp. 427-28.

Intellectual Property of Griffin Edward Oxford

other work of his provides a far more acceptable solution. The answer to the Liar Paradox does rest in part in the nature of self-reference, but we must first outline a correct account of a Proposition and better understand the rules which govern its use.

■■■

III. Possible World-States
The Answer We've Wanted All Along

In the pages that follow I shall attempt to outline a solution to the Liar Paradox using pragmatic concepts of truth and the theory of Possible World-States. The content of this theory is constructed from lectures given by Professor Richard Atkins in the class *Truth and Pragmatism* in the Fall of 2018 at Boston College. As there is very little literature regarding this theory, I will reference course notes and my own memory of discussions with Professor Atkins to build out the solution.[25]

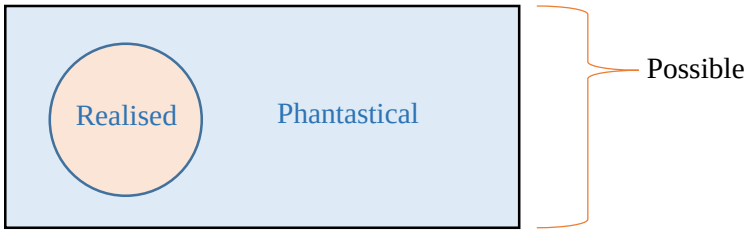
Up until this point, we have analyzed and rejected two attempts to solve the Liar Paradox. First, we discussed Tarski and his semantic theory – which falls short of a normative account and appears to exist in some ideal linguistically-logical world. Second, we turned to pragmatic theory and C.S. Peirce – which fails to use its own brilliant tools to arrive at a valid solution. Finally, we turn to Richard Atkins' view and Possible World-States – which comfortably handles issues of normativity, pragmatism, and the discomfort of the Liar Paradox. But what exactly do Possible World-States do for truth, and how does this theory aim to explain the world?

The universe exists – in one way or another[26]. There is some mind-independent reality, perhaps it is what we see, perhaps it is not. This exact existence – down to every last fiber, quark, byte, particle, or fundamental element – is the state of the world. We will call is the real, actual state the *Realised World-State*.

Due to the limitations of human perception, understanding, an inquiry, there are parts of this world-state which we do not know and parts which we cannot know. For example, we do not know whether the number of stars in the universe is odd or even, however one of these two options is *Realised*. For the sake of example, let us assume that the number of stars is even. Then we say that the world-state, §the number of stars in even§ is *Realised*. The symbolization '§X§' denotes that X is a world-state – this is not a proposition; it is a specific *order* (or *shape* as Atkins sometimes spoke of it) in which the universe exists. In contrast to the 'even' world-state, we say that the world-state, §the number of stars in the universe is odd§ is *phantastical*. That is, it is Phantasy, does not exist, is *un-Realised*. However, due to our current levels of observation, understanding, and inquiry, we do not know which of these world-states, in actuality, is *Realised*. Thus, we call these world-states *Possible World-States* – of which Realised World-States and Phantastical World-States are mutually exclusive subsets. This best understood in terms of the following graphic:

[25] Any flaws in reasoning, construction of the theory, or otherwise should be understood to be my own, and they are not entirely representative of Professor Atkin's view.
[26] Just ask Descartes.

Intellectual Property of Justin Edward Oxford



The set of Possible World-States is the entire box. As there are far more Phantastical World-States than Realised,[27] the Phantastical portion takes the majority of the space. The Realised World-States are then the smaller group of consistent world-states – that is, the Law of the Excluded Middle applies; $^{\S}X^{\S}$ and $^{\S}\neg X^{\S}$ cannot both be realised.[28] Thus far, there is no account of truth or falsity, only Realised and Phantastical World-States, and these terms (Truth/Realised; False/Phantastical) are not synonymous. A Realised World-State would be realised whether or not there existed intelligent life to make truth claims about it – as is case for Phantastical World-States. To understand the nature of truth, we must observe the nature of propositions in terms of World-State theory.

Atkins proposes two classes of propositions and one sub-class. The first of these classes is what he calls, L(ewisian)-Propositions. L-Propositions are abstract entities that supervene on Possible World-States. The second class is that of E-Propositions. These propositions are mostly conceptual, largely practical, and they are propositions which we use for various ends – to grasp reality, coordinate action, and *to understand truth*. However, E-Propositions only involve truth in a very specific context, under the tone of a certain attitude: when they are used to stand for reality *in the context of inquiry*. This specific context requires a sub-class of propositions – Atkins names these S-Propositions. These S-Propositions alone are the bearers of truth and falsity. We must take great care to understand this distinction.

E-Propositions can be understood in some sense as one's own predispositions or current understanding of the world. For example, an E-Propositions might be: "The experience at the TSA line is a pleasant experience," said in the context of someone asking what to expect on their first flight tomorrow morning, or, "That coffee is hot," in the context of one discerning whether or not to wait to drink their coffee. These propositions may be accurate – that is, the World-State which they describe may be realised – but there is no question of truth; they are assumed to be so, largely for pragmatic purposes. Truth does not arise until one questions one's own belief – or that belief is called into question, either by others or because of a realized misconception. Then arises the testable hypothesis: "My belief, B, *is true*" – an S-Proposition. From here, one can inquire about the world, and what was once an assumption is now a matter of truth or falsity. If one's beliefs are never tested, questioned, or otherwise contended, then one lives in a world of E-Propositions – a world without truth, because every conception is pre-accepted. Moreover, we avoid the issue that plagued Tarski in that we are able to understand propositions as being language-independent.[29]

The importance of the World-State / Proposition distinction is that we often wish to say that there is an independent, objective, real, *true* way that the world is, but at the same time we are limited in our understanding of this mind-independent world. The issue is that we are tempted to

[27] This should be apparent as one can have one way things *are*, and infinitely many ways they *could be*.
[28] There is some question of the symbolization here in regards to (¬), but it is not pivotal to our explanation. The key point for our purposes is that the world-state is CONSISTENT.
[29] Re-apply the desert island example. The thought in question is a proposition.

Intellectual Property of Lucas Edward Oxford

ascribe truth to both of these circumstances. In some sense we are tempted to say that truth is both absolute and fixed while also saying that it is in some way relativistic and can be changed or updated. We have a dualistic definition. However, viewing truth as having both of these properties in contradictory; it cannot be both fixed and changeable. Thus we distinguish World-States – the objective way the world is – with propositions – statements about the World-State which in some, specific classes can be true or false. In general, these S-Propositions (the bearers of truth and falsity in the context on inquiry) approach the Realised World-State. Whether or not they indeed can describe the Realised World-State exactly is dependent up one's opinion of the Limit Theory of Truth – i.e. whether it is the case that we approach Realised World-States, edging ever closer, but still they remain just out of our grasp (like the limit of 1 over X as X approaches zero), or whether it is the case that the end of inquiry is the thing itself. My own view is for the former, but that debate and proof is beyond the scope of this essay. Nonetheless, we thus have a theory of truth[30] which addresses the issue of normativity that plagues Tarski and also give us a rigorous account of the Proposition.

The question arises: just what *type of thing*, in the context of this theory, is the Liar Paradox? Is it a Realised or Phantastical World-State, an L-Proposition, an E-Proposition, an S-Proposition, or some other nonsense? The first answer one may be tempted to give is that the Liar Paradox is an S-Proposition. After all, it is talking of truth and falsity, it is attempting to point out the way that the world is, and we are using it in the context of inquiry – that is the entire purpose of this essay. Thus, this appears to be the proper classification. But while the Liar's statement is certainly forming a paradox in the realm of S-Propositions, we must first find assurance that paradox will not arise elsewhere – particularly because the World-State theory, in layman's terms, is a dualistic conceptualization of truth.[31]

To test this, let us consider our other options. First, if the Liar's statement is a World-State, then we must view it not in terms of truth propositionally, but in terms of objects realistically. That is, §This proposition is false§ is Realised or Phantastical. But regardless of the realm to which it is assigned, we are free from paradox. If it is Realised, then the proposition is a real part of the world, and if it is Phantastical, then it is phantasy – being true or false is not a quality which is possesses. It may be tempting to question further, but by questioning one is precisely moving from World-States to Propositions. For similar reasons, if the statement were an L-Proposition then we attain an identical result due to its supervenience onto the World-States – no paradox.

However, the Liar Paradox is a statement, in the world, about the world. Therefore, understanding the Liar's statement as being a world-state itself is improper. The world-state surrounding the statement is a state of opinion, understanding, and conception of an individual or group. The world-state is more appropriately stated as: §Some people hold that the statement: "This statement is false," is…§, and what follows is an explanation about the statements real consequences. Moreover, changing the Liar Paradox to be in terms of World-States, §This world-state is phantastical§ or something similar is an invalid formulation of a world-state. World-states are actual shapes of the universe and cannot be *about* anything at all.[32]

[30] Though, my explication should not be taken as an exhaustive account of Atkin's theory.
[31] Although, it distinguishes the definition into World-States and Propositions, and eliminates "truth-talk" from objective reality – the sentiment of normative truth still peeks through the blinds of the World-State language.
[32] This returns to the issue of (¬).

Intellectual Property of Tatum Edwards Oxford

The question remains as to whether the Liar's statement is or is not an E-Proposition, and further whether it is an S-Proposition. For if it is not an E-Proposition, then it cannot be an S-Proposition, it cannot bear truth or falsity, it is an invalid or absurd statement, and thus there is no paradox. And if it is an E-Proposition then we must look to the nature of S-Propositions and how the statement coheres to their nature – because it clearly appears to belong to that sub-class.

E-Propositions, at their base level, are the meanings of sentences, and expressions or descriptions of Possible World-States. In some way, they approximate what is believed to be reality, and they are intelligent life's attempt to articulate that belief. Sometimes these beliefs are useful, but sometime they are merely background concepts which exist subconsciously. From this, it is clear that to be an E-Proposition a statement must be about the world. Is the Liar's statement about the world in such a concrete way? I argue that it is not.

Consider the following statements, "The maxims of deflationist truth theory are false," and, "This statement is in English." Each of these statements makes some grab at the world[33], and each of them contains some aspect of the Liar's statement. The first statement can be traced to a real reference – the deflationist truth theory. The second statement references itself, and brings forward a testable fact – that it is in the English language. The Liar's statement, "This statement is false," only references itself; that is, it exists only in terms of itself and there is no conceivable, pragmatic way to apply it to any aspect of existence. Moreover, Atkins was correct in claiming that the, "Truth cannot be asserted of a proposition until the proposition has been enunciated. Once it has been enunciated, truth can be asserted of it by a rational agent..." and Peirce was correct by stating, "The object to which the subject of a proposition applies must be the object of a possible index… That is to say it must be real." The Liar's statement cannot reference a piece of the world because the piece of the world it seeks does not exist, and even if it did, it would be of so little consequence that it would not exist for any practical purpose.

Thus, it must be concluded that the statement, "This statement is false," makes no attempt to cohere to the world or to be a concept of it, and thus it is nonsense. Further, even if we accept the Liar's statement as an E-Proposition, it cannot be an S-Proposition as there is no subtext of inquiry. When asking about the Liar's statement, one is not attempting to get at the way that the world is. Instead, the statement is about nothing; it is not pointing to any feature of reality or understanding. It is equivalent to claiming that, "This cup of coffee is true," or, "This temperature is true." At bottom, if one rejects the account above and chooses to accept the Liar's sentence as an E-Proposition, then one is left with a meaningless E-Proposition.

IV. Closing Remarks

In the end we have come to see that Tarski, while solving the riddle in his own terms, ultimately failed because his terms were unrealistic. Further, Peirce, while edging closer to the solution, has his difficulties with self-reference. On both account difficulties were found because the Liar Paradox, when understood properly, is a statement about nothing, and it creates a problem only when one entertains it as a problem from the beginning. When understood in the context of a pragmatic, World-State model of Truth, the Liar Paradox is exposed for what it truly is: utter nonsense.

[33] Recall Peirce's notion of the *index*.

Intellectual Property of Justin Edward Oxford

Intellectual Property of Justin Edward Oxford

All works referenced are cited within in accordance with Chicago-Style.